Case Study on the Analysis of SDLC Models Applied to a Real-World Software Application

Aim

To analyze the Software Development Life Cycle (SDLC) models used in the development of a real-world software application and identify the most suitable SDLC model.

Objective

- Study different SDLC models.
- Analyze the development process of a real-world software application.
- Compare the advantages and limitations of various SDLC models.
- Recommend the most appropriate SDLC model for the application.

Selected Software Application

Netflix Streaming Platform

Overview

Netflix is a subscription-based video streaming platform that provides movies, television series, documentaries, and original content to millions of users worldwide.The platform allows users to:

• Register/Login

• Browse Movies and TV Shows

• Search Content

• Stream Videos

• Download Content

• Manage Profiles

• Receive Personalized Recommendations

• Rate and Review Content

Functional Requirements

User Module

• User Registration

• User Login

• Profile Management

• Subscription Management

• Content Search

• Video Streaming

• Download Content

Recommendation Module

• Analyze Viewing History

• Recommend Movies and Shows

• Personalized Suggestions

Content Management Module

• Upload New Content

• Update Content Information

• Manage Categories

Payment Module

• Subscription Payment

• Billing Management

• Payment Verification

Admin Module

• User Management

• Content Management

• Analytics and Reports

• Subscription Monitoring

SDLC Models Analysis

SDLC Model	Suitability	Advantages	Limitations
Waterfall	Low	Simple and structured	Difficult to accommodate frequent changes
Prototype	Medium	Better understanding of requirements	Increased development cost
Spiral	High	Excellent risk management	Complex and expensive
Agile	Very High	Continuous improvement and customer feedback	Requires experienced team
DevOps	Very High	Continuous Integration and Deployment	Infrastructure complexity

Why Agile is Best?

Reasons

• Customer requirements continuously evolve.

• New features are frequently introduced.

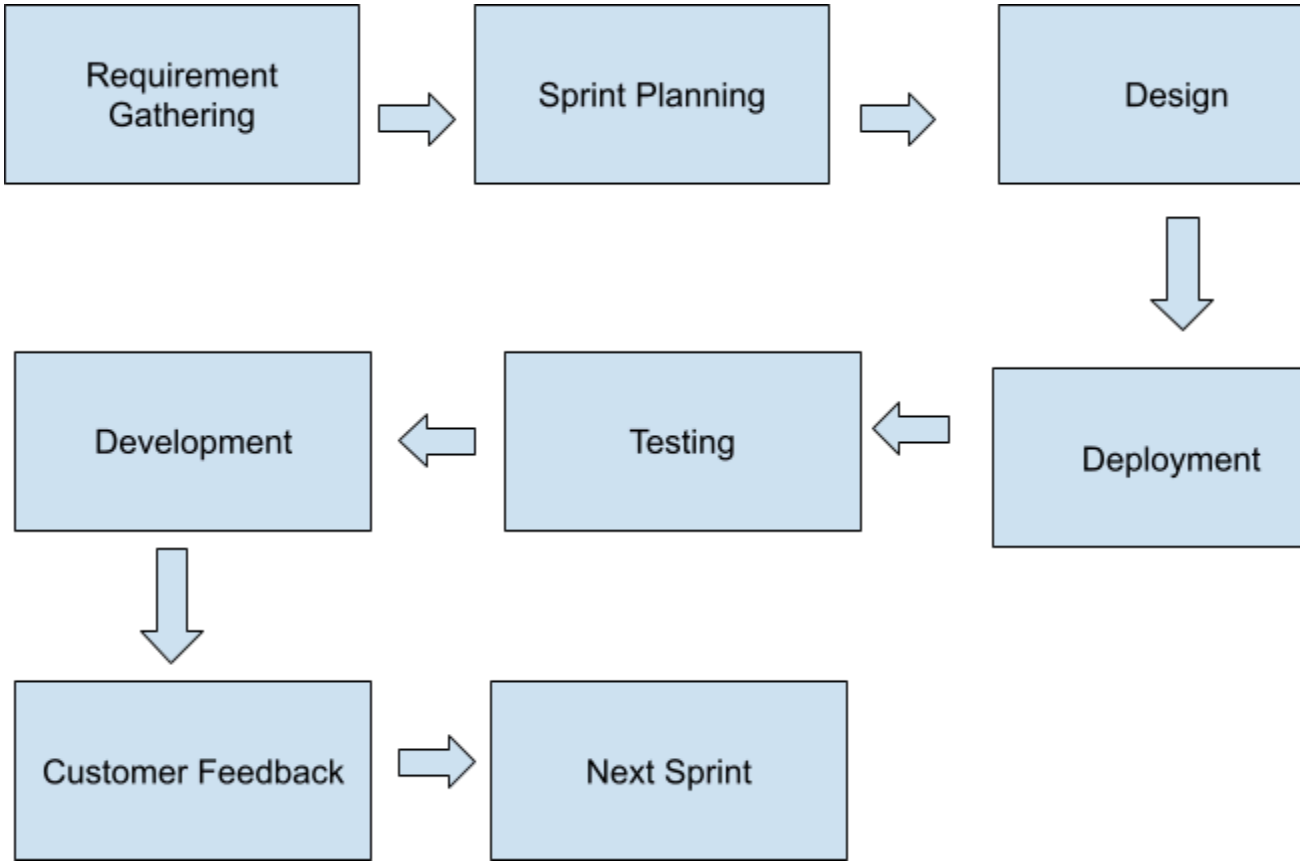
• User experience improvements are regularly implemented.

• Mobile and Smart TV applications require constant updates.

• Customer feedback helps improve recommendations and streaming quality.

• Faster release cycles are essential in a competitive streaming industry.

Agile Development Process



1. Requirement Analysis

Requirements Include:

• User Registration and Login

• Subscription Management

• Video Streaming

• Recommendation System

• Content Download

• Multi-device Support

Deliverables

• Software Requirement Specification (SRS)

• Requirement Analysis Report

2. System Design

Design Includes:

• Database Design

• User Interface Design

• Cloud Architecture Design

• API Design

• Recommendation Engine Design

Deliverables

• UML Diagrams

• Database Schema

• System Architecture Diagram

3. Development

Modules Developed:

• Authentication Module

• Streaming Module

• Recommendation Module

• Payment Module

• Content Management Module

Programming Languages and Technologies:

• Java

• Python

• JavaScript

• React

• Node.js

Database:

• MySQL

• PostgreSQL

• MongoDB

4. Testing

Testing Types:

• Unit Testing

• Integration Testing

• System Testing

• Performance Testing

• Security Testing

• User Acceptance Testing (UAT)

Example Test Cases

Test Case	Expected Result
Login	Successful Login
Invalid Password	Error Message
Video Playback	Video Streams Successfully
Subscription Payment	Payment Successful
Content Search	Relevant Results Displayed

5. Deployment

Deployment Options:

• Cloud Deployment

• Mobile Application Release

• Smart TV Application Deployment

• Web Platform Deployment

Cloud Platforms:

• AWS

• Microsoft Azure

• Google Cloud

6. Maintenance

Activities:

• Bug Fixes

• Security Updates

• Performance Optimization

• Feature Enhancements

• Recommendation Algorithm Improvements

• Infrastructure Scaling

Comparison of SDLC Models

Criteria	Waterfall	Prototype	Spiral	Agile	DevOps
Requirement Changes	Poor	Good	Excellent	Excellent	Excellent
Customer Feedback	Low	High	High	Very High	Very High
Risk Handling	Low	Medium	High	Medium	High
Delivery Speed	Slow	Medium	Medium	Fast	Very Fast

Cost	Low	Medium	High	Medium	Medium
Suitable for Netflix	No	Partially	Yes	Best	Best

Results

• Netflix requires continuous feature updates and rapid deployment.

• Agile SDLC supports iterative development and quick releases.

• Customer feedback can be incorporated continuously.

• DevOps enables continuous integration, testing, and deployment.

• Agile and DevOps together provide high scalability and reliability.

• The combination helps Netflix maintain a high-quality user experience.

Conclusion

The case study demonstrates that selecting an appropriate SDLC model depends on the nature of the software application. For dynamic platforms such as Netflix, Agile provides flexibility, faster development cycles, and continuous customer involvement. DevOps complements Agile by enabling automated testing, integration, deployment, and monitoring. Therefore, Agile combined with DevOps practices is the most suitable approach for developing and maintaining a large-scale streaming platform like Netflix.